

Transformation of Baseline Model into Single-Branch Network (SBNet)

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Abstract

This report presents the transformation of a baseline multimodal face-voice classification model into a Single-Branch Network (SBNet). The goal was to replace the dual-branch architecture with a shared representation model capable of learning from both modalities. The implementation required modifications to the model architecture, training procedure, and evaluation behavior. Experimental results show that while the SBNet is functional, it does not outperform the baseline, highlighting the challenges of unified multimodal learning.

Baseline Model

The baseline model consists of two independent branches: one for face embeddings and one for voice embeddings. Each branch processes its modality separately, and the resulting representations are combined using a linear fusion layer. The fused representation is then passed to a classifier. This architecture allows the model to learn modality-specific features and combine them effectively, resulting in strong performance in both multimodal and cross-lingual settings.

Modifications for SBNet

To implement the Single-Branch Network, several key changes were made. First, the separate face and voice branches were removed and replaced by a shared network. Since the original feature dimensions differ (face: 512, audio: 192), linear projection layers were introduced to map both modalities into a common embedding space. Second, the fusion layer was removed and replaced by a shared representation computed through the same neural network. Third, the training procedure was modified to follow a sequential strategy: each batch is processed twice, once with audio-only input and once with face-only input. This ensures that both modalities update the same network parameters. Finally, the forward pass was updated to handle missing modalities by dynamically selecting the available representation or averaging both when present.

Results

Setting	Baseline	SBNet
Seen (0% missing)	93.86%	52.63%
Seen (100% missing)	76.90%	62.57%
Unseen (0% missing)	91.38%	26.72%
Unseen (100% missing)	74.57%	46.98%

Analysis

The SBNet model shows significantly lower performance compared to the baseline. A key observation is that accuracy improves when the face modality is removed, indicating that the model

relies heavily on audio features and does not effectively integrate visual information. This behavior suggests that the shared representation is not sufficient to capture complementary modality features. Additionally, performance drops drastically in the unseen language setting, highlighting limited generalization.

Conclusion

The transformation from a dual-branch model to a Single-Branch Network was successfully implemented. However, the results show that this architecture struggles to match the performance of the baseline. The findings emphasize the importance of modality-specific learning and effective fusion mechanisms. Future improvements could include hybrid architectures or better regularization techniques to balance modality contributions.